

Father's Day Crossword Solutions

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1 6	2 4
3 3	2 2

ONE ACROSS

Let $s = a + b + c$, $q = ab + bc + ca$, and $r = abc$. We are given $s = 14$. Also,

$$(a - b)^2 + (b - c)^2 + (c - a)^2 = 2(a^2 + b^2 + c^2 - ab - bc - ca).$$

Since $a^2 + b^2 + c^2 = s^2 - 2q$, we get $32 = 2(s^2 - 3q)$. Thus, $16 = s^2 - 3q = 196 - 3q$, so $q = 60$.

Now observe that $a + b - c = s - 2c$, $b + c - a = s - 2a$, and $c + a - b = s - 2b$. Since $s = 14$, we have

$$(a + b - c)(b + c - a)(c + a - b) = (14 - 2a)(14 - 2b)(14 - 2c).$$

Factoring out 2 from each term gives

$$(14 - 2a)(14 - 2b)(14 - 2c) = 8(7 - a)(7 - b)(7 - c).$$

Therefore, we get

$$8(7 - a)(7 - b)(7 - c) = 104 \Rightarrow (7 - a)(7 - b)(7 - c) = 13.$$

Expanding,

$$(7 - a)(7 - b)(7 - c) = 343 - 49(a + b + c) + 7(ab + bc + ca) - abc.$$

Substitute $s = 14$ and $q = 60$ to get $13 = 343 - 49(14) + 7(60) - r \Rightarrow 13 = 343 - 686 + 420 - r = 77 - r$, so $r = \boxed{64}$.

ONE DOWN

Note that the modulus M is a product of 6 distinct odd primes.

For any odd prime p , the congruence $x^2 \equiv 1 \pmod{p}$ has exactly two solutions: $x \equiv \pm 1 \pmod{p}$. To prove this, note that the given congruence can be written as $x^2 - 1 \equiv 0 \Rightarrow (x - 1)(x + 1) \equiv 0 \pmod{p}$. This means that $p \mid (x - 1)(x + 1)$. Since p is prime, we must then have either $p \mid x - 1$ or $p \mid x + 1$, which is equivalent to saying $x \equiv \pm 1 \pmod{p}$.

Thus, for each of the 6 prime factors of M , we have two choices for the residue, and by the Chinese Remainder Theorem, the total number of solutions is $2^6 = 64$. One of these is the code, which must be excluded, so the number of reversible codes is $64 - 1 = \boxed{63}$.

TWO DOWN

Let C_n be the number of ways to pair $2n$ points on a circle with noncrossing chords. We need C_5 .

Choose one fixed point. Suppose it is paired with another point so that there are $2k$ points on one side of the chord and $2n - 2 - 2k$ points on the other side. The pairings on the two sides are independent, so

$$C_n = \sum_{k=0}^{n-1} C_k C_{n-1-k}.$$

Also, $C_0 = 1$. This is exactly the Catalan recurrence, so

$$C_n = \frac{1}{n+1} \binom{2n}{n}.$$

For $n = 5$,

$$C_5 = \frac{1}{6} \binom{10}{5} = \frac{252}{6} = \boxed{42}.$$

THREE ACROSS

Let O be the midpoint of the common chord PQ . Since PQ is perpendicular to AC , points A, O, C are collinear.

Let $AO = x$. Then, $CO = 42 - x$.

Because P lies on both circles, triangle AOP has hypotenuse $AP = 20$, while triangle COP has hypotenuse $CP = 34$. Thus, $OP^2 = 20^2 - x^2$ and $OP^2 = 34^2 - (42 - x)^2$. Therefore, $20^2 - x^2 = 34^2 - (42 - x)^2$. Expanding, we get

$$400 - x^2 = 1156 - (1764 - 84x + x^2),$$

so

$$400 - x^2 = -608 + 84x - x^2.$$

Hence, $84x = 1008$, and therefore $x = 12$.

Now, $OP^2 = 20^2 - 12^2 = 400 - 144 = 256$, so $OP = 16$. Since O is the midpoint of PQ , $PQ = 2OP = \boxed{32}$.